Flow Calibration

Understanding flow:

Although a 2 mm nozzle tip is recommended for printing spirulina paste, it is possible to print paths that are significantly wider than the nozzle diameter. This can be achieved by increasing the FLOW (%) value in the Settings menu, which controls the amount of material extruded through the nozzle at a given deposition speed (**Figure 1**).



Figure 1: Flow influence on material deposition

3D printing typically deposit material one layer at a time. However, for cup shaped geometries, instead of printing the cup in successive layers, vase mode deposition strategy enables the cup wall to be printed as one continuous line, reducing print time, eliminating seam marks, and producing a smoother, more consistent extrusion path.

A correct FLOW(%) allows the model to be 3D printed without gaps or excessive material and allows for the cup walls to be printed at the desired thickness (deposition width) when using vase mode.

The deposition width value can be adjusted in **Settings → Deposition Width** (mm). Changing this value will modify the nozzle path and therefore the generated G-code file used to run the 3D printer.
It is therefore important to check the flow rate to ensure that the deposited 3D printed line width matches the expected deposition width value.

A flow rate that is too low will result in under-extrusion:

The deposited line will be narrower than intended, which will lead to gaps, weak base/walls, poor layer bonding and unstable geometry (**Figure 2**).

A flow rate that is too high will result in over-extrusion:

The deposited line will be wider than intended, causing material build-up, loss of detail, deformation of the print, and poor surface quality.



Figure 2: Example of under-extrusion

As spirulina paste is typically prepared in small batches using basic measuring equipment and by different operators, **variations in the recipe ingredient ratios will occur**. These variations will directly influence how the paste flows through the nozzle.

Default printing parameters are provided based on the recommended paste recipe. However, depending on the final paste consistency, **adjustment of the flow rate might be required** to achieve the desired deposition width.

Flow Calibration

How to run the flow test?

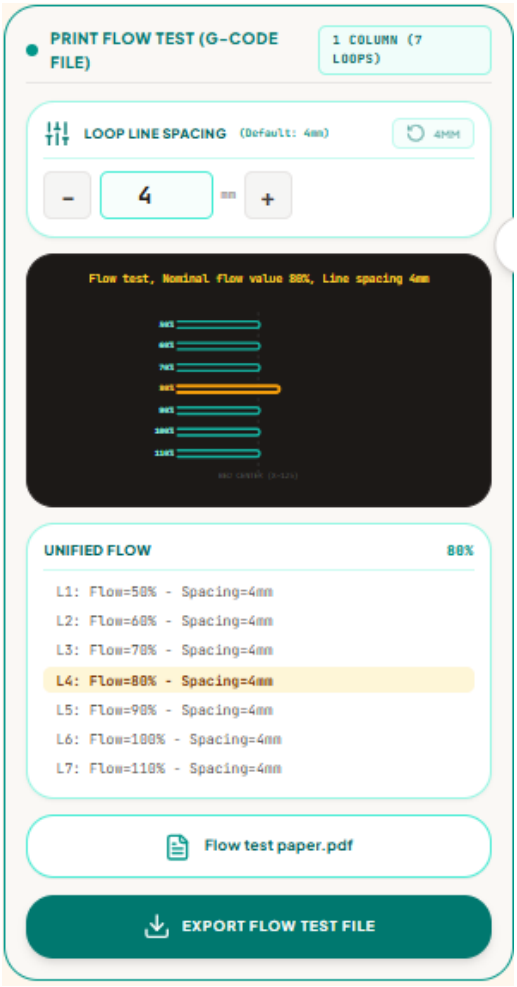


Figure 3: Print flow test in the app

To help determine the optimum flow rate, a dedicated flow test is available in the app (**Save → Print Flow Test**). The flow test file can be saved as a G-code file and printed on your 3D printer (**Figure 3**).

The flow test consists of printing seven return loops: one loop at the nominal flow rate, three loops at progressively lower flow rates (10% reduction at each step), and three loops at progressively higher flow rates (10% increase at each step) (**Figure 4**).

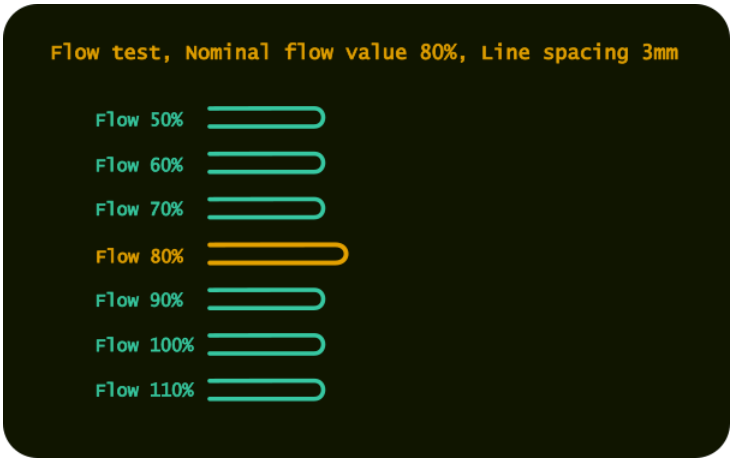


Figure 4: Print flow test layout

Flow Calibration

How to check and correct the flow value?

The correct flow rate is the one for which the two lines meet without leaving a visible gap, while avoiding excessive overlap or material build-up (**Figure 5**).

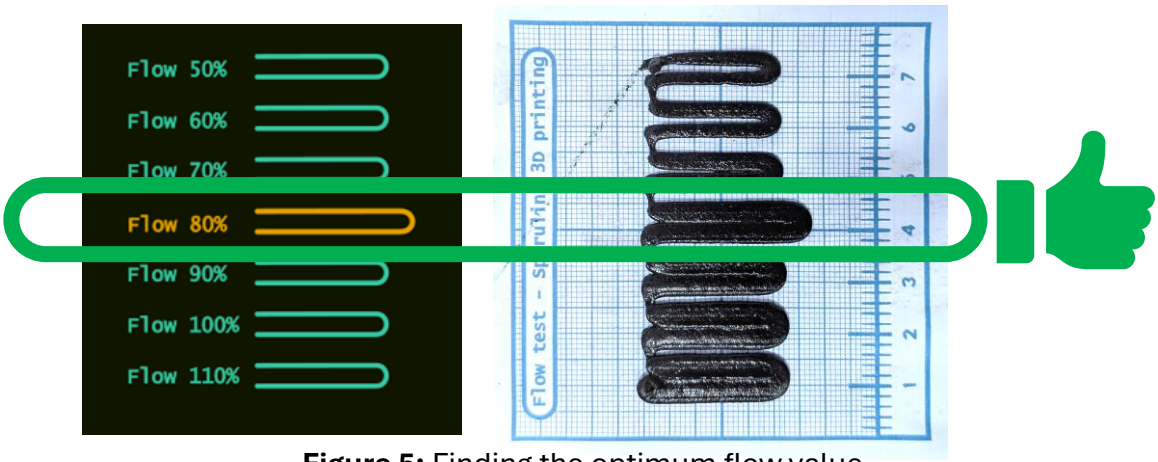


Figure 5: Finding the optimum flow value

Make a note of the corresponding flow rate value. If this value differs from the nominal flow rate, **update the FLOW (%) value in the Settings menu** and **click Apply New Settings.** Any G-code files subsequently generated by the app will use the updated flow rate value.

Note that the settings are not permanently saved. If the app is reloaded or opened in a new browser window, all settings will revert to their default values.

- If all loops show closed gaps and signs of **over-extrusion, decrease the FLOW (%)** value in the Settings menu and repeat the flow calibration test.
- If all loops show visible **gaps between the return lines, increase the FLOW (%)** value in the Settings menu and repeat the flow calibration test.
- If **gaps remain despite increasing FLOW (%), reduce the SPEED** value in the Settings menu and repeat the flow calibration test.
- If the **extruder motor clicks or skips steps**: the material is difficult to extrude. **Reduce the SPEED** value in the Settings menu and repeat the flow calibration test.
- If the extruder **motor continues to click or skip steps** after reducing speed, the paste is likely too thick. Dilute the paste slightly or, preferably, **prepare a new batch** following the recipe and preparation method carefully.